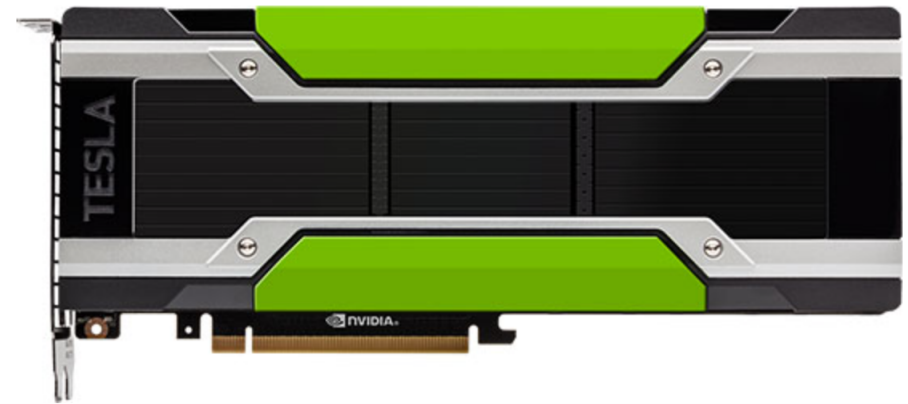


# Lecture Unit 10.2

## GPUs in HPC

# GPU connectivity

- The CPU and GPU are separate components with separate memory
- Connection is often via PCI Express (PCI-e) interface
- Industry standard connection: connect to any machine with enough power, cooling and physical space
- PCI-e has a relatively high latency which can impact performance

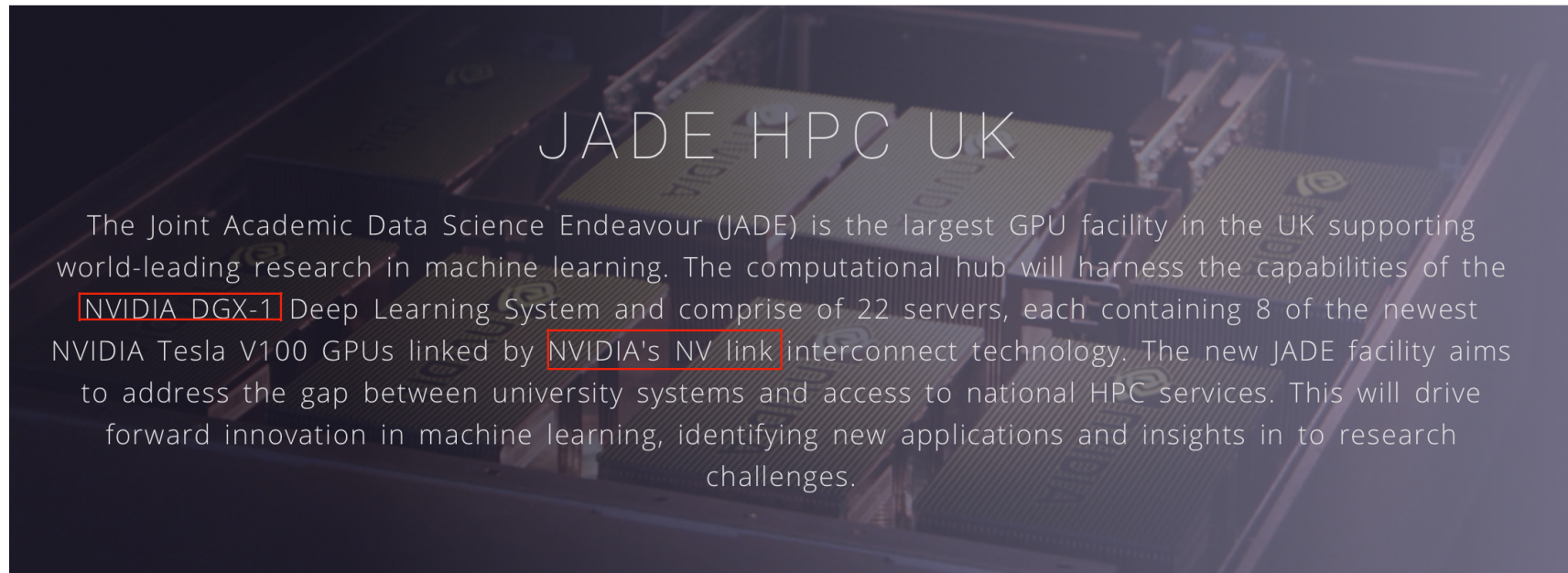


<http://www.nvidia.co.uk/object/tesla-p100-uk.html>

→ Minimise transfer of data between host and device

# Dense GPU systems

<http://www.jade.ac.uk>



NVIDIA's proprietary "NV link" technology provides better connectivity than PCI-e

# Energy efficiency

- Energy efficiency is an important factor in HPC system design (power consumption and cooling requirements)
- GPUs offer good energy efficiency
- The Green500 list is a companion to the Top500 list which measures energy efficiency (GFlops/Watt)

# Green 500 November 2024

<https://top500.org/lists/green500/2024/11/>

| Rank | TOP500 Rank | System                                                                                                                                                                                                                             | Cores  | Rmax (PFlop/s) | Power (kW) | Energy Efficiency (GFlops/watts) |
|------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------|------------|----------------------------------|
| 1    | 222         | JEDI - BullSequana XH3000, Grace Hopper Superchip 72C 3GHz, NVIDIA GH200 Superchip, Quad-Rail NVIDIA InfiniBand NDR200, ParTec/EVIDEN EuroHPC/FZJ Germany                                                                          | 19,584 | 4.50           | 67         | 72.733                           |
| 2    | 122         | ROMEO-2025 - BullSequana XH3000, Grace Hopper Superchip 72C 3GHz, NVIDIA GH200 Superchip, Quad-Rail NVIDIA InfiniBand NDR200, Red Hat Enterprise Linux, EVIDEN ROMEO HPC Center - Champagne-Ardenne France                         | 47,328 | 9.86           | 160        | 70.912                           |
| 3    | 440         | Adastra 2 - HPE Cray EX255a, AMD 4th Gen EPYC 24C 1.8GHz, AMD Instinct MI300A, Slingshot-11, RHEL, HPE Grand Equipement National de Calcul Intensif - Centre Informatique National de l'Enseignement Suprieur (GENCI-CINES) France | 16,128 | 2.53           | 37         | 69.098                           |
| 4    | 155         | Isambard-AI phase 1 - HPE Cray EX254n, NVIDIA Grace 72C 3.1GHz, NVIDIA GH200 Superchip, Slingshot-11, HPE University of Bristol United Kingdom                                                                                     | 34,272 | 7.42           | 117        | 68.835                           |
| 5    | 51          | Capella - Lenovo ThinkSystem SD665-N V3, AMD EPYC 9334 32C 2.7GHz, Nvidia H100 SXM5 94Gb Infiniband NDR200, AlmaLinux 9.4, MEGWARE TU Dresden, ZIH Germany                                                                         | 85,248 | 24.06          | 445        | 68.053                           |

El Capitan is not in the top10 of the Green500 list (ranked 18)

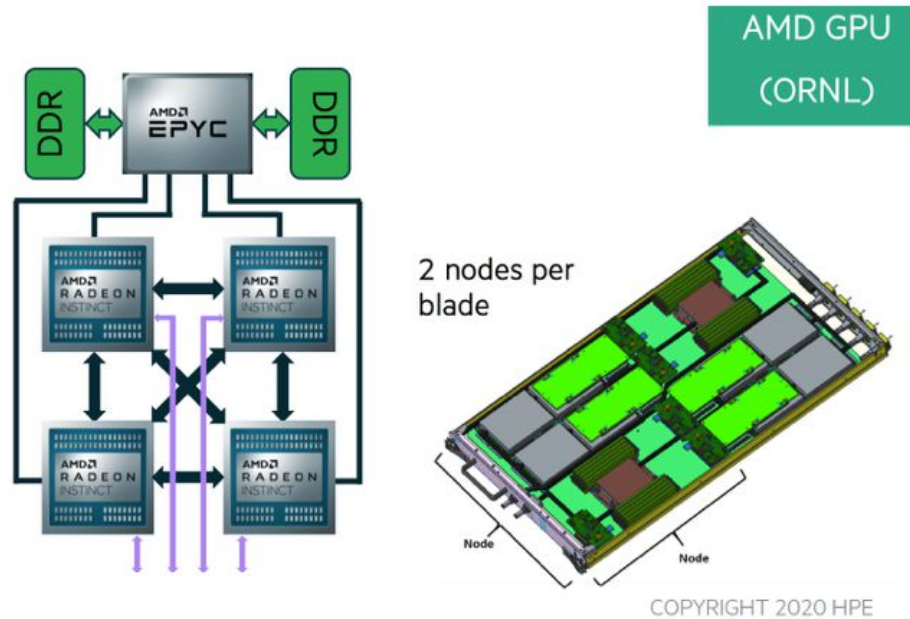
# The exascale era

- [Frontier](#): AMD EPYC CPUs and AMD Radeon GPUs (the first exascale supercomputer)
- [Fugaku](#): A64fx ARM CPUs
- [El Capitan](#): AMD EPYC CPUs and AMD Radeon GPUs
- [Aurora](#): Intel Xeon CPUs and Intel Xe GPUs
- [Perlmutter](#): AMD EPYC CPUs and NVIDIA GPUs
- [JUPITER](#): NVIDIA GPUs (Europe's first exascale supercomputer)

# Hardware diversity

- In recent years Intel CPUs and NVIDIA GPUs have dominated the market but this is changing:
  - CPUs from AMD, ARM and Intel
  - GPUs from ARM, Intel and NVIDIA
- Portability will be an important consideration

# Frontier has an accelerator-centric architecture:



Interconnect

<https://www.olcf.ornl.gov/frontier/#4>



# Unit summary

- GPUs are often connected via the industry standard PCI-e interface
- PCI-e has a relatively high latency so data transfer between the host and the device needs to be minimised
- GPU accelerated HPC systems can deliver good energy efficiency as measured by the Green500 list
- Most of the planned exascale system use GPUs
- These GPUs are from different vendors so we need to consider portability